GGG-F-325b
October 31, 1966
SUPERSEDING
Fed. Spec. GGG-F-325a
February 17, 1960

FEDERAL SPECIFICATION

FILE, HAND (AMERICAN PATTERN) AND RASP, HAND

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers American pattern hand files and rasps.

1.1.1 *Federal specification coverage.* Federal specifications do not cover all varieties of the commodity indicated by the titles of the specifications but include only those generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types, classes, and styles.* American Pattern files and rasps shall be of the following types, classes, and styles as specified (see 6.2).

Files:
- Type:
  - I—Aluminum.
    - Class 1—Flat.
    - Class 2—Half-round.
  - II—Bandsaw, blunt.
    - Class 1—Regular.
    - Class 2—Slim.
  - IV—Cantsaw.
  - V—Crosscut ("Great American").
  - VI—Double-ender.
  - VII—Flat.
    - Style A—Bastard-cut.
    - Style B—Second-cut.
    - Style C—Smooth-cut.
    - Style D—Single-cut with chip-breaking grooves.
  - IX—Half-round.
    - Style A—Bastard-cut.
    - Style B—Second-cut.
    - Style C—Smooth-cut.
  - X—Hand.
    - Class 2—Regular.
      - Style A—Bastard-cut.
      - Style B—Second-cut.
      - Style C—Smooth-cut.
  - XI—Knife.
    - Style A—Bastard-cut.
    - Style B—Second-cut.
    - Style C—Smooth-cut.
  - XII—Lead-float.
    - Class 1—Flat.
    - Class 2—Half-round.
  - XIII—Mill.
    - Class 1—Blunt (special crosscut).
    - Class 2—Regular.
      - Style A—Bastard-cut.
      - Style B—Second-cut.
      - Style C—Smooth-cut.
    - Class 3—Two round edges.
    - Class 4—One round edge.
  - XIV—Pillar.
    - Style A—Bastard-cut.
    - Style B—Second-cut.
    - Style C—Smooth-cut.
  - XVI—Round.
    - Class 1—Regular.
      - Style A—Bastard-cut.
      - Style B—Second-cut.
      - Style C—Smooth-cut.
    - Class 2—Chain saw.
  - XVII—Square.
    - Style A—Bastard-cut.
    - Style B—Second-cut.
    - Style C—Smooth-cut.
  - XVIII—Taper.
    - Class 1—Regular.

FSC 5110

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:39Z
Check the source to verify that this is the current version before use.

Class 2—Slim.
Class 3—Extra-slim.
Class 4—Double-extra-slim.

XIX—Three square.
Style A—Bastard-cut.
Style B—Second-cut.
Style C—Smooth-cut.

XX—Warding.
Style A—Bastard-cut.
Style B—Second-cut.
Style C—Smooth-cut.

XXII—Long angle lathe.

XXIII—Milled curved teeth, flat flexible blade.
Style A—Standard-cut.
Style B—Fine-cut.
Style C—Smooth-cut.

Rasps:

Type:

XXIV—Cabinet.
Style B—Second-cut.

XXV—Horse.
Class 1—Plain, regular.
Class 2—Plain, slim.
Class 3—Tanged, regular.
Class 4—Tanged, slim.

XXVI—Shoe, half-round.

XXVII—Wood.
Class 1—Flat.
Style A—Bastard-cut.
Class 2—Half-round.
Style A—Bastard-cut.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

QQ-S-634—Steel, Bar, Carbon, Cold Finished, (Standard Quality).

Federal Standard:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-H-15424—Hand Tools, Packaging of.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Qualification.** The files and rasps furnished under this specification shall be a product which has been tested, passed the qualification tests specified herein, and listed on, or approved for listing on, the applicable Qualified Products List (see 6.3).

3.2 **Illustrations.** The illustrations shown herein are descriptive, and not restrictive, and are included for the convenience of requisitioning and purchasing officers and manufacturers, and are not intended to preclude the purchase of tools which are otherwise in accordance with this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:39Z
Check the source to verify that this is the current version before use.

3.3 Files. The height and form of teeth and the overall file shape of each individual type, class, style, cut, and length shall have the characteristics of commercial American Pattern files.

3.4 Rasps. The height and form of the teeth and the overall rasp shape of each individual type, class, style, cut, and length shall have the characteristics of conventional commercial rasps. File teeth, if any, shall be American Pattern.

3.5 Teeth.

3.5.1 *File teeth.* Teeth of any individual surface or edge shall be uniformly cut (or etched) and of uniform height throughout the fully cut portion. Teeth shall be of the proper hardness and sharp enough to pass the tests in section 4. There shall be no missing or deformed teeth. Teeth shall be free from decarburization and extreme brittleness, which might cause breakage under normal service conditions.

3.5.1.1 *Cuts.*

3.5.1.1.1 *Bastard-cut second-cut, and smooth-cut.* The terms "bastard-cut", "second-cut", and "smooth-cut" are the indexes of the relative number of teeth per unit length. On the average, in files of a given length, bastard-cut files have fewer teeth per unit length than do second-cut files; smooth-cut files have more.

3.5.1.1.2 *Single-cut and double-cut.* The terms "single-cut" and "double-cut" denote the number of independent sets of parallel cuts of teeth on a file surface.

3.5.1.1.3 Upcut teeth of double-cut file surfaces are the primary cutting teeth. Overcut teeth are considered to be superimposed on the upcut teeth although in manufacturing practice, the overcut teeth are generally cut first.

3.5.2 *Rasp teeth.* Rasp teeth shall be formed by raising portions of the metal from the surface of the blank. The teeth shall be entirely separate from one another and the top of each tooth shall be rounded. Teeth shall be uniformly cut and of uniform height throughout. The tolerances for rasp teeth shall be plus or minus 1 tooth per row and one row per inch.

3.6 Single-cut and double-cut surfaces. A single-cut surface, where specified, shall have one set of parallel cuts on the surface. A double-cut surface, where specified, shall have two sets of parallel cuts on the surface, one set crossing the other.

3.7 Number of teeth per inch. The number of teeth per inch shall be the number of parallel lines of cuts counted on the surface in a distance of one inch parallel with the longitudinal centerline of the surface.

3.8 Tangs. With the exception of types VI and XXIII, files: type XXV, classes 1 and 2; and type XXVI, rasps, each file and rasp shall be provided with a substantial tang of conventional shape and size suitable for attachment to a wood handle.

3.9 Length. The length of types VI and XXIII, files; type XXV, classes 1 and 2; and type XXVI, rasps, shall be the overall length. The length of all other files and rasps shall be measured from the point (free or front end) to the heel (junction of the tang with the body of the file or rasp). The length shall be as specified (see 6.2), and shall be not less than that shown in the applicable table and not more than 10 percent greater, except that for type XXIII, the length shall be plus or minus 3/16 inch.

3.10 Cross section. The cross section of each individual type, class, and style shall conform to the applicable figure.

3.11 Material. Files and rasps shall be of steel or a chemical composition which, together with the shape, height, and method of cutting (or etching) of the teeth and subsequent hardening by heat treatment, results in the production of files and rasps having good cutting qualities.

3.12 Heat treatment. The cutting surfaces of all files and rasps shall be heat treated

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:39Z
Check the source to verify that this is the current version before use.

TABLE I. *Type I, aluuminum, class* 1, *flat*

Length	Dimensions at largest section		Number of teeth per inch		Allowable (maximum)
			Minimum	Maximum	
	A	B			
Inches	*Inches*	*Inch*			*Degrees*
6	5/8	5/32	16	18	3
8	13/16	7/32	14	16	3
10	1	1/4	14	16	3
12	1- 3/16	17/64	14	16	3

and hardened to a uniform hardness capable of passing the test requirements in section 4. The tang shall be tempered to a hardness not greater than 30, nor less than 26 Rockwell C scale at a point approximately midway of its length and width.

3.13 **Condition.** Files and rasps shall be clean and free from rust and scale.

3.14 **Marking.** Each file and rasp shall be stamped in a plain and permanent manner with the name of the manufacturer, or with a trademark of such known character that the source of manufacturer may be readily determined.

3.15 **Tolerances.** The tolerances on width and thicknesses shall be as follows:

Files (except warding)	Tolerance, inch	
	Width	Thickness or diameter
Length, 6 inches and under	± 1/16	± 1/32
Length, over 6 inches	± 1/16	± 1/16
Warding files (lengths 4 to 10 inches)	± 1/16	± 1/16
Rasps (lengths 6 to 14 inches	± 1/8	± 1/8

3.16 Type I, aluminum.

3.16.1 *Class* 1, *flat.* The files shall be of uniform thickness from heel to point and shall taper in width at least one-third of the length to the point, shall be double-cut on each flat, and shall have two square edges with each edge single-cut. The files shall conform to table I and shall be similar to figure 1.



FIGURE 1. Type I, aluminum, class 1, flat.

3.16.2 *Class* 2, *half-round.* The files shall taper in both width and thickness at least one-third of the length to the point: the flat side shall be double-cut, and the back side (convex side) shall be single-cut or double-cut, at the option of the manufacturer. The files shall conform to table II and shall be similar to figure 2.



FIGURE 2. Type I, aluminum, class 2, half-round.

TABLE II. *Type I, aluminum, class* 2, *half-round*

Length	Dimensions at largest section		Number of teeth per inch				Allowable warpage (maximum)
			Back		Flat		
			Minimum	Maximum	Minimum	Maximum	
	A	B					
Inches	*Inches*	*Inch*					*Degrees*
6	5/8	5/32	16	18	16	18	2
8	3/4	7/32	15	17	15	17	2
10	15/16	9/32	15	17	15	17	2
12	1- 1/8	11/32	14	16	14	16	2

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:39Z
Check the source to verify that this is the current version before use.

TABLE III. *Type II, bandsaw, blunt, class 1, regular.*

Length	Dimensions of section A	Number of teeth per inch				Allowable warpage (maximum)
		Sides		Edges		
		Minimum	Maximum	Minimum	Maximum	
Inches	*Inch*					*Degrees*
6	1/2	40	45	50	60	2
8	5/8	36	40	45	55	2

3.17 Type II, bandsaw, blunt.

3.17.1 *Class 1, regular.* All sides shall be of equal width from heel to point; all edges shall be straight and parallel from heel to point, and shall be single-cut on all sides. Edges shall be rounded and set (blunted before and after the surface cuts) and then cut with at least two parallel rows of teeth so as to be suitable for filing rounded gullets of bandsaw teeth. The files shall conform to table III and shall be similar to figure 3.

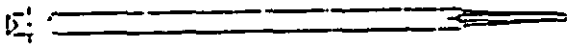


FIGURE 3. Type II, bandsaw, blunt, classes 1 and 2, regular and slim.

3.17.2 *Class 2, slim.* The files shall conform to 3.17.1 and table IV and shall be similar to figure 3.

3.18 Type IV, cantsaw. The files shall be of uniform width and thickness from heel to point; the two sides shall be of equal width from heel to point and at equal angles with the back; all edges shall be straight and parallel from heel to point. Files shall be single-cut on back and sides. Narrow edges or sides shall be square with back and shall be single-cut. The files shall conform to table V and shall be similar to figure 4.

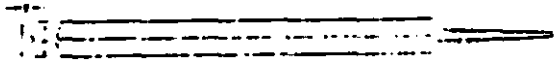


FIGURE 4. Type IV, cantsaw.

TABLE IV. *Type II, bandsaw, blunt, class 2, slim.*

Length	Dimensions of section A	Number of teeth per inch				Allowable warpage (maximum)
		Sides		Edges		
		Maximum	Minimum	Minimum	Maximum	
Inches	*Inch*					*Degrees*
6	3/8	55	50	60	65	2
8	1/2	45	41	50	55	2

TABLE V. *Type IV, cantsaw.*

Length	Dimensions of cross section		Number of teeth per inch				Allowable warpage (maximum)
			Back		Sides		
	A	B	Minimum	Maximum	Minimum	Maximum	
Inches	*Inch*	*Inch*					*Degrees*
6	17/32	13/64	49	55	49	55	2
8	11/16	17/64	44	48	44	48	2
10	13/16	5/16	39	43	39	43	2

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:39Z
Check the source to verify that this is the current version before use.

TABLE VI. *Type V, crosscut ("Great American").*

Length	Dimensions of cross section		Number of teeth per inch		Allowable warpage (maximum)
	A	B	Minimum	Maximum	
Inches	*Inch*	*Inch*			*Degrees*
8	11/16	9/32	42	48	3
10	13/16	23/64	38	43	3

3.19 **Type V, crosscut ("Great American").** The file shall be of uniform width and thickness from heel to point. The two sides shall be of equal width from heel to point and at equal angles with the median longitudinal plane of the file and all corners shall be straight and parallel from heel to point. The back shall be rounded for filing gullets. Files shall be single-cut on back, sides, and bottom edge. The files shall conform to table VI and shall be similar to figure 5.

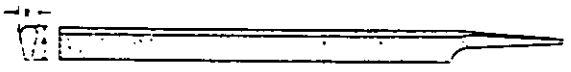


FIGURE 5. Type V, crosscut ("Great American")

3.20 **Type VI, double-ender.** The files shall taper on all sides from the center of the file to each point, and all sides at any point on the file shall be of equal width. Files shall be single-cut on each side. Edges shall be set (blunted before and after the cut), and then single-cut so as to be suitable for filing gullets between saw teeth. Type VI files shall conform to table VII and shall be similar to figure 6.

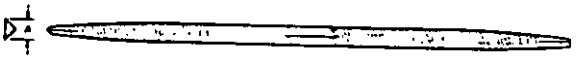


FIGURE 6. Type VI, double-ender.

TABLE VII. *Type VI, double-ender*

Length	Dimensions at largest section	Number of teeth per inch	
	A	Minimum	Maximum
Inches	*Inch*		
6	8/16	58	63
7	7/32	58	63
8	9/32	56	59
9	11/32	52	56
10	3/8	47	51

3.21 **Type VII, flat, styles A, B, C, and D; bastard-cut, second-cut, smooth-cut, and single-cut with chip-breaking grooves.** Files shall be of uniform width and thickness from heel to midpoint (approx.). Files shall taper in width from approximately the midpoint to the point. Files shall be either parallel in thickness, or shall taper slightly in thickness from approximately the middle of the length to the point. Styles A, B, and C shall be double-cut on each flat. Style D shall be single-cut on each flat and have from four to six chip-breaking grooves per inch, cut diagonally, and designed for fast stock removal while producing a reaonably smooth surface. The chip-breaking grooves shall run from left-to-right, or right-to-left, at the manufacturer's option. All four styles shall have two square edges with each edge single-cut. The files shall conform to table VIII and shall be similar to figure 7.



FIGURE 7. Type VII, flat, styles A, B, C, and D; bastard-cut, second-cut, smooth-cut, and single-cut with chip-breaking grooves.

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:39Z
Check the source to verify that this is the current version before use.

TABLE VIII. *Type VII, flat, styles A, B, C, and D*

Length	Dimensions at largest section		Number of teeth per inch		Allowable distortion (maximum)	Allowable warpage (maximum)
	A	B	Maximum	Minimum		
Inches	*Inches*	*Inch*			*Inch*	*Degrees*
Style A. bastard-cut						
4	7/16	7/64	40	45	0.006	2
6	5/8	5/32	30	33	.008	2
8	13/16	7/32	23	26	.010	2
10	1	1/4	20	23	.025	2
12	1- 3/16	9/32	17	19	.025	2
14	1- 3/8	5/16	16	18	.050	2
16	1-17/32	11/32	14	16	.060	2
18	1-23/32	3/8	13	15	.070	2
Style B. second-cut						
4	7/16	7/64	50	58	0.006	2
6	5/8	5/32	41	45	.008	2
8	13/16	7/32	33	37	.010	2
10	1	1/4	29	32	.025	2
12	1- 3/16	9/32	26	28	.025	2
14	1- 3/8	5/16	23	25	.050	2
16	1-17/32	11/32	21	23	.060	2
Style C. smooth-cut						
4	7/16	7/64	65	73	0.006	2
6	5/8	5/32	52	57	.008	2
8	13/16	7/32	44	48	.010	2
10	1	1/4	37	41	.025	2
12	1- 3/16	9/32	33	36	.025	2
14	1- 3/8	5/16	29	32	.050	2
16	1-17/32	11/32	29	32	.060	2
Style D. single-cut with chip-breaking grooves						
8	25/32	7/32	20	26	0.010	2
10	31/32	1/4	18	22	.025	2
12	1- 5/32	9/32	17	19	.025	2
14	1-11/32	5/16	16	18	.050	2

3.22 Type IX, half-round, styles A, B, and C; bastard-cut, second-cut and smooth-cut. The file shall taper in width at least one-third of the length to the point. Files shall be either parallel in thickness or shall taper slightly in thickness from approximately the middle of the length to the point. The flat side shall be double-cut and the back side (convex side) shall be double-cut, except that style B, second-cut files in 4- and 6-inch lengths and all style C, smooth-cut files shall be single-cut, or double-cut, at the option of the manufacturer. Type IX files shall conform to table IX and shall be similar to figure 8.



FIGURE 8. Type IX, half-round, styles A, B, and C; bastard-cut, second-cut, and smooth-cut.

3.23 Type X, hand, class 2, regular, styles A, B, and C; bastard-cut, second-cut, and smooth-cut. The files shall be of uniform width from heel to point. Files shall be either parallel in thickness or shall taper slightly in thickness from approximately the middle of the length to the point. They shall be double-cut on each flat, and shall have two square edges with one edge single-

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:39Z
Check the source to verify that this is the current version before use.

TABLE IX. *Type IX, half-round, styles A, B, and C*

Length	Dimensions at largest section		Number of teeth per inch				Allowable warpage (maximum)
			Back		Flat		
	A	B	Minimum	Maximum	Minimum	Maximum	
Inches	*Inches*	*Inch*					*Degrees*
Style A. bastard-cut							
4	7/16	1/8	37	45	37	45	3
6	5/8	5/32	29	34	29	34	3
8	3/4	7/32	24	28	24	28	3
10	15/16	9/32	21	24	21	24	3
12	1- 1/8	11/32	19	21	18	20	3
14	1- 5/16	13/32	17	19	16	18	3
16	1- 1/2	29/64	15	17	14	16	3
Style B. second-cut							
4	7/16	1/8	51	59	51	59	3
6	5/8	5/32	41	46	41	46	3
8	3/4	7/32	33	37	33	37	3
10	15/16	9/32	29	32	29	32	3
12	1- 1/8	11/32	26	29	26	29	3
14	1- 5/16	13/32	23	26	23	26	3
Style C. smooth-cut							
4	7/16	1/8	65	73	65	73	3
6	5/8	5/32	50	56	50	56	3
8	3/4	7/32	44	50	44	50	3
10	15/16	9/32	39	43	39	43	3
12	1- 1/8	11/32	34	37	34	37	3
14	1- 5/16	13/32	31	34	31	34	3
16	1- 1/2	29/64	29	32	29	32	3

cut and the other edge smooth (safe). The files shall conform to table X and shall be similar to figure 9.



FIGURE 9. Type X, hand, class 2, regular, styles A, B, and C; bastard-cut, second-cut, and smooth-cut.

3.24 Type XI, knife, styles A, B, and C; bastard-cut, second-cut, and smooth-cut. The files shall be of uniform thickness or tapered slightly in thickness toward the point, at the manufacturer's option, and shall taper in width at least one-third of the length to the point. Files shall have a knife cross section with an included angle between sides of 9 to 15°, inclusive. Files shall be double-cut on each wide flat and single-cut on the thin edge. The back shall be single-cut or smooth (safe), at the option of the contractor. The files shall conform to table XI and shall be similar to figure 10.



FIGURE 10. Type XI, knife, styles A, B, and C; bastard-cut, second-cut, and smooth-cut.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:39Z
Check the source to verify that this is the current version before use.

TABLE X. *Type X, hand, class 2, regular, styles A, B, and C*

Length	Dimensions at largest section		Number of teeth per inch		Allowable distortion (maximum)	Allowable warpage (maximum)
			Maximum	Minimum		
	A	B				
Inches	*Inches*	*Inch*			*Inch*	*Degrees*
Style A. bastard-cut						
6	5/8	5/32	31	35	0.008	2
8	13/16	7/32	23	26	.010	2
10	1	1/4	20	23	.025	2
12	1- 3/16	17/64	17	19	.025	2
14	1- 3/8	19/64	16	18	.050	2
Style B. second-cut						
6	5/16	5/32	41	46	0.008	2
8	13/16	7/32	34	38	.010	2
10	1	1/4	29	32	.025	2
12	1- 3/16	17/64	26	28	.025	2
Style C. smooth-cut						
6	5/8	5/32	50	56	0.008	2
8	13/16	7/32	44	50	.010	2
10	1	1/4	38	42	.025	2
12	1- 3/16	17/64	34	38	.025	2
14	1- 3/8	19/64	31	34	.050	2

TABLE XI. *Type XI, knife, styles A, B, and C*

Length	Dimensions at largest section			Number of teeth per inch		Allowable warpage (maximum)
				Minimum	Maximum	
	A	B	C			
Inches	*Inch*	*Inch*	*Inches*			*Degrees*
Style A. bastard-cut						
4	7/64	1/32	15/32	49	55	3
6	5/32	1/16	11/16	41	45	5
8	3/16	1/16	7/8	33	37	6
10	1/4	1/16	1- 1/16	28	32	7
Style B. second-cut						
4	7/64	1/32	15/32	60	68	3
6	5/32	1/16	11/16	52	59	5
8	3/16	1/16	7/8	42	48	6
10	1/4	1/16	1- 1/16	34	39	7
Style C. smooth-cut						
4	7/64	1/32	15/32	82	90	3
6	5/32	1/16	11/16	66	73	5
8	3/16	1/16	7/8	54	60	6
10	1/4	1/16	1- 1/16	45	49	7

3.25 Type XII, lead-float.

3.25.1 *Class 1, flat.* The files shall be of uniform width and thickness from heel to approximate middle of length and shall taper in both width and thickness from approximate middle to the point of file or the thickness may be uniform throughout, at the manufacturer's option. The file shall be single-cut on each flat, and shall have two

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:39Z
Check the source to verify that this is the current version before use.

TABLE XII. *Type XII, lead-float, class 1, flat.*

Length	Dimensions at largest section		Number of teeth per inch		Allowable distortion (maximum)	Allowable warpage (maximum)
			Minimum	Maximum		
	A	B				
Inches	*Inches*	*Inch*			*Inch*	*Degrees*
10	1	1/4	15	17	0.015	2
12	1-3/16	17/64	13	15	.020	3

square edges with each edge single-cut. Teeth shall be suitable for filing soft metal such as lead and babbitt. The files shall conform to table XII and shall be similar to figure 11.

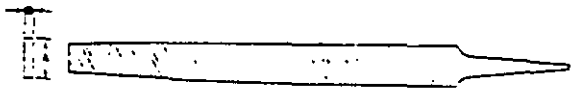


FIGURE 11. Type XII, lead-float, class 1, flat.

3.25.2 *Class 2, half-round.* The files shall be of uniform width and thickness from heel to approximate middle of length, shall taper in both width and thickness from the approximate middle to point of the file, and shall be single-cut on each surface. The files shall conform to table XIII and shall be similar to figure 12.

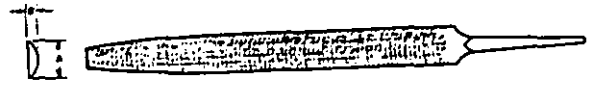


FIGURE 12. Type XII, lead-float, class 2, half-round.

3.26 **Type XIII, mill.** Unless otherwise specified (see 6.2), the file shall be single-cut on each flat.

3.26.1 *Class 1, blunt (special crosscut).* The files shall be of uniform width and thickness from heel to point and shall have two square edges with each edge single-cut. The files shall conform to table XIV and shall be similar to figure 13.

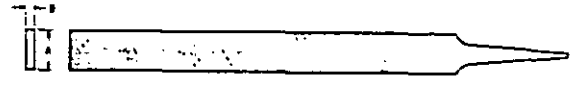


FIGURE 13. Type XIII, mill, class 1, blunt, (special crosscut).

TABLE XIII. *Type XII, lead-float, class 2, half-round*

Length	Dimensions at largest section		Number of teeth per inch				Allowable warpage (maximum)
			Back		Flat		
			Minimum	Maximum	Minimum	Maximum	
	A	B					
Inches	*Inches*	*Inch*					*Degrees*
10	15/16	9/32	15	17	15	17	3
12	1- 1/8	11/32	13	15	13	15	3

TABLE XIV. *Type XXII, mill, class 1, blunt, (special crosscut)*

Length	Dimensions at largest section		Number of teeth per inch				Allowable distortion (maximum)	Allowable warpage (maximum)
			Flats		Edges			
			Min.	Max.	Min.	Max.		
	A	B						
Inches	*Inch*	*Inch*					*Inch*	*Degrees*
6	5/8	1/8	48	52	44	59	0.008	2
7	45/64	1/8	46	50	42	54	.008	2
8	13/16	5/32	43	48	42	50	.010	2
10	1	3/16	38	42	38	44	.015	2

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:39Z
Check the source to verify that this is the current version before use.

TABLE XV. *Type XIII, mill, class 2, regular, styles A, B, and C*

Length	Dimensions at largest section		Number of teeth per inch				Allowable distortion (maximum)	Allowable warpage (maximum)
			Flats		Edges			
	A	B	Min.	Max.	Min.	Max.		
Inches	*Inches*	*Inch*					*Inch*	*Degrees*
Style A. bastard-cut								
4	7/16	5/64	54	62	56	70	0.006	2
6	5/8	1/8	48	52	44	59	.008	2
7	11/16	1/8	46	50	42	52	.008	2
8	13/16	5/32	43	46	42	48	.010	2
10	1	3/16	38	41	38	44	.015	2
12	1- 3/16	7/32	34	37	33	38	.015	2
14	1-11/32	1/4	30	33	31	34	.020	2
16	1-17/32	9/32	26	29	27	30	.020	2
Style B. second-cut								
4	7/16	5/64	66	70	64	72	0.006	2
6	5/8	1/8	59	63	61	70	.008	2
8	13/16	5/32	50	53	52	61	.010	2
10	1	3/16	44	47	46	54	.015	2
12	1- 3/16	7/32	39	42	41	48	.015	2
14	1-11/32	1/4	34	37	36	43	.020	2
Style C. smooth-cut								
6	5/8	1/8	68	74	72	81	0.008	2
8	13/16	5/32	56	62	60	71	.010	2
10	1	3/16	51	56	54	64	.015	2
12	1- 3/16	7/32	45	50	48	57	.015	2
14	1-11/32	1/4	39	43	42	49	.020	2

3.26.2 *Class 2, regular, styles A, B, and C; bastard-cut, second-cut, and smooth-cut.* The files shall be of uniform thickness or tapered slightly in thickness from heel to point. Files shall taper in width at least one-third of the length to the point and shall have two square edges with each edge single-cut. The files shall conform to table XV and shall be similar to figure 14.

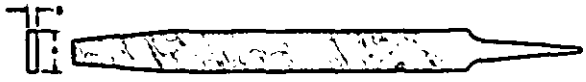


FIGURE 14. Type XIII, mill, class 2, regular styles A, B, and C; bastard-cut, second-cut, and smooth-cut.

3.26.3 *Class 3, two round edges.* The files shall be bastard-cut. They shall be identical with class 2, style A files except they shall have semicircular edges with each edge single-cut. The files shall conform to table XVI and shall be similar to figure 15.

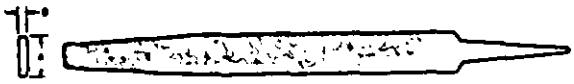


FIGURE 15. Type XIII, mill, class 3, two round edges.

TABLE XVI. *Type XIII, mill, class 3, two round edges*

Length	Dimensions at largest section		Number of teeth per inch				Allowable distortion (maximum)	Allowable warpage (maximum)
			Flats		Edges			
	A	B	Min.	Max.	Min.	Max.		
Inches	*Inch*	*Inch*					*Inch*	*Degrees*
8	13/16	5/32	43	46	42	48	0.010	2
10	1	3/16	38	41	38	44	.015	2

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:39Z
Check the source to verify that this is the current version before use.

TABLE XVII, *Type XIV, pillar, styles A, B, and C*

Length	Dimensions at largest section		Number of teeth per inch		Allowable distortion (maximum)	Allowable warpage (maximum)
Inches	A	B	Minimum	Maximum		
	Inch	*Inch*			*Inch*	*Degrees*
			Style A. bastard-cut			
6	7/16	7/32	34	40	0.008	2
8	9/16	9/32	28	32	.010	2
10	11/16	11/32	23	26	.025	2
12	13/16	13/32	20	23	.025	2
			Style B. second-cut			
6	7/16	7/32	44	50	0.008	2
8	9/16	9/32	37	42	.010	2
10	11/16	11/32	33	37	.025	2
			Style C. smooth-cut			
6	7/16	7/32	56	64	0.008	2
8	9/16	9/32	47	54	.010	2

3.26.4 *Class 4, one round edge.* Class 4 files shall be furnished in accordance with the dimensions of table XVI except they shall have one round single cut edge and one square single cut edge.

3.27 **Type XIV, pillar styles A, B, and C; bastard-cut, second-cut, and smooth-cut.** The files shall be of uniform width from heel to point of rectangular cross section and shall taper slightly in thickness toward the point. The proportion of width to thickness shall be approximately two-to-one. Files shall be double-cut on each flat and have two square edges with one single-cut edge and one safe edge. The files shall conform to table XVII and shall be similar to figure 16.



FIGURE 16. Type XIV, pillar, styles A, B, and C; bastard-cut, second-cut, and smooth-cut.

3.28 **Type XVI, round.**

3.28.1 *Class 1 regular, styles A, B, and C; bastard-cut, second-cut, and smooth-cut.* The files shall taper at least one-third of the length to point. Bastard-cut files shall be double-cut except that the 4-inch length may be single-cut. Second-cut files shall be single-cut except the 12- and 14-inch lengths which shall be double-cut. Smooth-cut files shall be single-cut. The files shall conform to table XVIII and shall be similar to figure 17.



FIGURE 17. Type XVI, round, class 1, regular, styles A, B, and C; bastard-cut, second-cut, and smooth-cut.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:39Z
Check the source to verify that this is the current version before use.

TABLE XVIII. *Type XVI, round, class 1, regular, styles A, B, and C*

Length	Diameter at largest section	Number of teeth per inch		Allowable distortion (maximum)
		Minimum	Maximum	
Inches	A			
	Inch			*Inch*
Style A. bastard-cut				
4	3/16	37	44	0.010
6	1/4	26	31	.010
7	17/64	25	30	.015
8	5/16	21	25	.020
10	3/8	19	21	.025
12	1/2	16	19	.025
14	5/8	15	18	.030
16	3/4	15	17	.040
Style B. second-cut				
4	3/16	46	53	0.010
6	1/4	34	39	.010
8	5/16	28	32	.020
10	3/8	26	29	.025
12	1/2	23	26	.025
14	5/8	21	23	.030
Style C. smooth-cut				
4	3/16	58	65	0.010
6	1/4	46	51	.010
8	5/16	39	43	.020
10	3/8	36	39	.025
12	1/2	32	35	.025
14	5/8	30	33	.030

3.28.2 *Class 2, chain saw.* The files shall be round and parallel throughout their length. Files shall be single-cut or double-cut. The teeth may extend to the point or terminate a short distance back to provide a safe point. The files shall conform to table XIX and shall be similar to figure 18.



FIGURE 18. Type XVI, round, class 2, chain saw

TABLE XIX. *Type XVI, round, class 2, chain saw*

Length	Diameter	Number of teeth per inch		Allowable distortion (maximum)
		Minimum	Maximum	
	A			
Inches	*Inch*			*Inch*
8	3/16	48	66	0.020
8	7/32	48	66	.020
8	1/4	48	66	.020
8	9/32	48	66	.020
8	5/16	48	66	.020
8	3/8	48	66	.020

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:39Z
Check the source to verify that this is the current version before use.

TABLE XX. *Type XVII, square, styles A, B, and C*

Length	Dimensions at largest section	Number of teeth per inch		Allowable distortion (maximum)
		Minimum	Maximum	
Inches	A *Inch*			*Inch*
Style A. bastard-cut				
4	3/16	43	48	0.010
6	1/4	33	37	.015
8	5/16	25	29	.020
10	3/8	22	25	.020
12	1/2	19	21	.020
14	5/8	16	18	.025
16	3/4	15	17	.025
Style B. second-cut				
4	3/16	56	63	0.010
6	1/4	43	49	.015
8	5/16	36	41	.020
10	3/8	31	35	.020
12	1/2	29	31	.020
14	5/8	26	28	.025
Style C. smooth-cut				
4	3/16	72	79	0.010
6	1/4	55	61	.015
8	5/16	44	49	.020
10	3/8	39	44	.020
12	1/2	36	39	.020
14	5/8	33	36	.025

3.29 Type XVII, square, styles A, B, and C; bastard-cut, second-cut, and smooth-cut. Files shall taper in both dimensions (width and thickness) at least one-third of the cut length from the point. Files shall be double-cut on all surfaces, conform to table XX, and shall be similar to figure 19.



FIGURE 19. Type XVII, square, styles A, B, and C; bastard-cut, second-cut, and smooth-cut.

3.30 Type XVIII, taper. The files shall taper on all sides at least one-third of the length to the point and be single-cut on each surface. The sides at any point on the file shall be of equal width. Edges shall be set (blunted) and then single-cut. No portion of the tapered length shall extend outside of the area of the largest cross section of the file.

3.30.1 *Class 1, regular.* Files shall conform to table XXI and shall be similar to figure 20.

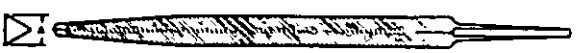


FIGURE 20. Type XVIII, taper, class 1, regular.

TABLE XXI. *Type XVIII, taper, class 1, regular*

Length	Dimensions at largest section	Number of teeth per inch			
		Surfaces		Edges	
		Minimum	Maximum	Minimum	Maximum
Inches	A *Inch*				
6	1/2	40	45	51	56
7	9/16	37	42	45	51
8	5/8	36	40	45	51
10	3/4	32	36	42	46

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:39Z
Check the source to verify that this is the current version before use.

TABLE XXII. *Type XVIII, taper, class 2, slim*

Length	Dimensions at largest section	Number of teeth per inch			
		Surfaces		Edges	
		Minimum	Maximum	Minimum	Maximum
	A				
Inches	*Inch*				
4	1/4	58	63	67	72
5	5/16	54	59	61	66
6	3/8	50	55	60	66
7	13/32	45	50	51	56
8	1/2	41	45	46	55
10	19/32	37	42	44	50

3.30.2 *Class 2, slim.* The files shall conform to table XXII and shall be similar to figure 21.

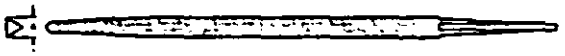


FIGURE 21. Type XVIII, taper, class 2, slim.

3.30.3 *Class 3, extra-slim.* The files shall conform to table XXIII and shall be similar to figure 22.

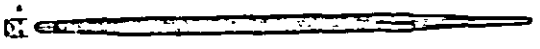


FIGURE 22. Type XVIII, taper, class 3, extra-slim.

3.30.4 *Class 4, double-extra-slim.* The files shall conform to table XXIV and shall be similar to figure 23.

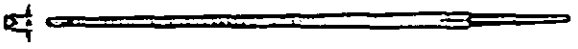


FIGURE 23. Type XVIII, taper, class 4, double-extra-slim.

3.31 Type XIX, three-square, styles A, B, and C; bastard-cut, second-cut, and smooth-cut. The files shall be double-cut on each surface, taper on all sides at least one-third of the length to the point, and each side at a given point shall be of equal width.

TABLE XXIII. *Type XVIII, taper, class 3, extra-slim*

Length	Dimensions at largest section	Number of teeth per inch			
		Surfaces		Edges	
		Minimum	Maximum	Minimum	Maximum
	A				
Inches	*Inch*				
4	3/16	61	68	72	79
5	7/32	59	64	68	73
6	9/32	55	60	62	70
7	11/32	51	56	57	65
8	13/32	48	52	54	60

TABLE XXIV. *Type XVIII, taper, class 4, double-extra-slim*

Length	Dimensions at largest section	Number of teeth per inch			
		Surfaces		Edges	
		Minimum	Maximum	Minimum	Maximum
	A				
Inches	*Inch*				
4	5/32	62	70	73	81
5	3/16	60	66	70	77
6	7/32	58	63	66	71
7	1/4	55	60	62	70
8	5/16	52	57	58	65

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:39Z
Check the source to verify that this is the current version before use.

Edges shall be set and cut so as to be suitable for filing internal angles and clearing out corners. No portion of the tapered length shall extend outside of the area of the largest cross section of the file. The files shall conform to table XXV and shall be similar to figure 24.



FIGURE 24. Type XIX, three-square, styles A, B, and C; bastard-cut, second-cut, and smooth-cut.

TABLE XXV. *Type XIX three-square, styles A, B, and C*

Length	Dimensions at largest section	Number of teeth per inch	
		Sides	
		Minimum	Maximum
	A		
Inches	*Inch*		
Style A. bastard-cut			
6	1/2	33	37
8	5/8	25	29
10	3/4	22	25
12	27/32	18	22
Style B. second-cut			
6	1/2	43	49
8	5/8	36	41
10	3/4	31	35
Style C. smooth-cut			
6	1/2	55	63
8	5/8	44	49
10	3/4	39	44

TABLE XXVI. *Type XX, warding, styles A, B, and C*

Length	Dimensions at largest section		Number of teeth per inch		Allowable distortion (maximum)	Allowable warpage (maximum)
			Minimum	Maximum		
	A	B				
Inches	*Inch*	*Inch*			*Inch*	*Degrees*
Style A. bastard-cut						
4	7/16	1/16	50	55	0.005	5
6	5/8	5/64	41	45	.008	5
8	25/32	3/32	34	37	.010	5
10	15/16	1/8	28	31	.015	7
Style B. second-cut						
4	7/16	1/16	68	78	0.005	5
6	5/8	5/64	55	59	.008	5
8	25/32	3/32	44	48	.010	5
10	15/16	1/8	36	39	.015	7
Style C. smooth-cut						
4	7/16	1/16	84	91	0.005	5
6	5/8	5/64	68	73	.008	5
8	25/32	3/32	55	60	.010	5
10	15/16	1/8	43	48	.015	7

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:39Z
Check the source to verify that this is the current version before use.

TABLE XXVII. *Type XXII, long angle lathe*

Length	Dimensions at largest section		Number of teeth per inch		Allowable distortion (maximum)	Allowable warpage (maximum)
	A	B	Minimum	Maximum		
Inches	*Inches*	*Inch*			*Inch*	*Degrees*
10	1	1/4	22	26	0.015	2
12	1- 3/16	17/64	20	24	.017	2
14	1-23/64	19/64	19	21	.020	2

3.32 Type XX, warding, styles A, B, and C; bastard-cut, second-cut, and smooth-cut. The files shall be of uniform thickness throughout and shall taper in width from heel to point with relatively sharp taper toward the point. Files shall be double-cut on each flat and shall have two squared edges with each edge single-cut. The files shall conform to table XXVI and shall be similar to figure 25.



FIGURE 25. Type XX, warding, styles A, B, and C; bastard-cut, second-cut, and smooth-cut.

3.33 Type XXII, long angle lathe. Long angle lathe files shall be of uniform width and thickness from heel to approximate middle of length and shall taper in both width and thickness from approximate middle to the point, or the thickness may be uniform throughout, at the manufacturer's option. These files shall have single-cut faces with teeth placed at a long angle. The edges shall be safe (uncut). These files shall conform to table XVII and shall be similar to figure 26.

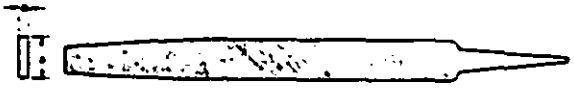


FIGURE 26. Type XXII, long angle lathe.

TABLE XXVIII. *Type XXIII, milled curved teeth, flat, flexible blade, styles A, B, and C*

Length	Dimensions at largest section		Number of teeth per inch		Allowable distortion (maximum)	Allowable warpage (maximum)	Size of holes (approximate)	Distance between holes (approximate)
	A	B	Minimum	Maximum				
Inches	*Inches*	*Inch*			*Inch*	*Degrees*	*Inch*	*Inches*
Style A. standard-cut								
8	25/32	5/32	13	15	0.015	3	3/16	6- 3/4
10	1	5/32	11	13	.025	3	3/16	8- 3/4
12	1- 5/32	3/16	9	11	.025	3	1/4	10-13/16
14	1-11/32	3/16	7	9	.050	3	1/4	12-13/16
Style B. fine-cut								
8	25/32	5/32	15	17	.015	3	3/16	6- 3/4
10	1	5/32	13	15	.025	3	3/16	8- 3/4
12	1- 5/32	3/16	12	14	.025	3	1/4	10-13/16
14	1-11/32	3/16	11	13	.050	3	1/4	12-13/16
Style C. smooth-cut								
8	25/32	5/32	17	19	.015	3	3/16	6- 3/4
10	1	5/32	16	18	.025	3	3/16	8- 3/4
12	1- 5/32	3/16	15	17	.025	3	1/4	10-13/16
14	1-11/32	3/16	14	16	.050	3	1/4	12-13/16

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:39Z
Check the source to verify that this is the current version before use.

TABLE XXIX. *Type XXIV, rasps, cabinet, style B*

Length	Dimensions at largest section		Teeth				Allowable warpage (maximum)
			Back		Flat		
	A	B	Teeth per row	Rows per inch	Teeth per row	Rows per inch	
Inches	*Inches*	*Inch*					*Degrees*
B, second-cut.							
6	25/32	3/16	7	9	6	9	8
8	15/16	1/4	9	8-2/3	8	8-2/3	8
10	1- 1/8	9/32	11	6-2/3	10	6-2/3	8
12	1- 5/16	11/32	11	6-2/3	10	6-2/3	8
14	1- 9/16	13/32	12	6-2/3	11	6-2/3	8

3.34 Type XXIII, milled curved teeth, flat, flexible blade, styles A, B, and C; standard-cut, fine-cut, and smooth-cut. The files shall be of uniform width and thickness throughout their length. Both flat sides shall have milled curved teeth and both edges shall be uncut. The corners and edges at each end shall be beveled in accordance with good manufacturing practices. The files shall be capable of being flexed safely up to 1/8-inch arc for 8-inch files; 1/4-inch for 10-inch files; 3/8-inch for 12-inch files; and 5/8-inch for 14-inch files. Holes countersunk 82° on both sides of the file, shall be provided near each end for attaching to a standard file holder. The files shall conform to table XXVIII and shall be similar to figure 27.

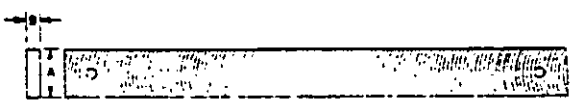


FIGURE 27. Type XXIII, milled curved teeth, flat, flexible blade, styles A, B, and C; standard-cut, fine-cut, and smooth-cut.

3.35 Type XXIV, rasps, cabinet, style B, second-cut. Rasps shall taper in both width and thickness toward the point and shall have two edges with each edge single-cut. The rasps shall conform to table XXIX and shall be similar to figure 28.



FIGURE 28. Type XXIV, rasps, cabinet, style B, second-cut.

3.36 Type XXV, rasps, horse. The rasps shall have rasp teeth on one side and file teeth on the other side. File teeth shall be double-cut and shall have two square edges with each edge single-cut.

3.36.1 ***Classes 1 and 2, plain regular, and plain, slim.*** The rasps shall be of uniform width and thickness over the entire length and shall be of double-end construction. Rasps shall conform to table XXX and shall be similar to figure 29.

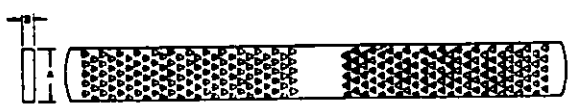


FIGURE 29. Type XXV, rasps, horse, classes 1 and 2; plain, regular, and plain, slim.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:39Z
Check the source to verify that this is the current version before use.

TABLE XXX. *Type XXV, rasps, horse, classes 1 and 2*

Length	Dimensions of sections		Rasp side		Number of teeth per inch		Allowable warpage (maximum)
			Teeth per row	Rows per inch	File side		
	A	B			Maximum	Minimum	
Inches	*Inches*	*Inch*					*Degrees*
Class 1. plain, regular							
12	1- 9/32	11/32	5	7	11	15	8
14	1-17/32	3/8	5	7	10	14	8
16	1-23/32	7/16	5	7	10	13	8
Class 2. plain, slim							
18	1-19/32	13/32	5	7	10	13	8

3.36.2 *Classes 3 and 4, tanged, regular, and tanged, slim.* The rasps shall be of uniform width and thickness from heel to point, conform to table XXXI, and shall be similar to figure 30.



FIGURE 30. Type XXV, rasps, horse, classes 3 and 4; tanged, regular, and tanged, slim.

3.37 Type XXVI, rasps, shoe, half-round. Rasps shall be of uniform width and thickness over the entire length and shall be of double-end construction. One-half of the rasp shall have rasp teeth on front and back: the other half, double-cut file teeth, front and back. Rasps shall have two edges square with the flat face and the edges shall be uncut (safe). The rasps shall conform to table XXXII and shall be similar to figure 31.



FIGURE 31. Type XXVI, rasps, shoe, half-round.

3.38 Type XXVII, rasps, wood.

3.38.1 *Class 1, flat, style A, bastard-cut.* The rasps shall taper in width from heel to point and shall taper in thickness from heel to point, or shall be of uniform thickness throughout, at the option of the manufacturer. Rasps shall have rasp teeth on each surface and two square edges with each edge

TABLE XXXI. *Type XXV, rasps, horse, classes 3 and 4*

Length	Dimensions of sections		Rasp side		Number of teeth per inch		Allowable warpage (maximum)
			Teeth per row	Rows per inch	File side		
	A	B			Minimum	Maximum	
Inches	*Inches*	*Inch*					*Degrees*
Class 3. tanged, regular							
14	1-23/32	7/16	5	7	10	13	8
16	1- 7/8	15/32	5	7	10	13	8
Class 4. tanged, slim							
14	1-23/32	23/64	6	7	10	14	8

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:39Z
Check the source to verify that this is the current version before use.

TABLE XXXII. *Type XXVI, rasps, shoe, half-round*

Length	Dimensions at section		Teeth								Allowable warpage (maximum)
			Rasp end				File end teeth per inch				
			Flat		Back		Back		Flat		
			Teeth per row	Rows per inch	Teeth per row	Rows per inch	Min.	Max.	Min.	Max.	
	A	B									
Inches	*Inches*	*Inch*									*Degrees*
8	7/8	7/32	8	12	9	12	19	21	18	21	8
9	31/32	1/4	8	12	9	12	18	20	11	20	8
10	1-1/16	9/32	8	11	9	11	17	20	10	20	8

single-cut. The rasps shall conform to table XXXIII and shall be similar to figure 32.



FIGURE 32. Type XXVII, rasps, wood, class 1, flat, style A; bastard-cut.

3.38.2 *Class 2, half-round, style A: bastard-cut.* Rasps shall taper in both width and thickness from the heel to the point. The rasps shall conform to table XXXIV and shall be similar to figure 33.



FIGURE 33. Type XXVII, rasps, wood, class 2, half-round, style A; bastard-cut.

TABLE XXXIII. *Type XXVII, rasps, wood, class 1, flat, style A*

Length	Dimensions at largest section		Teeth		Allowable distortion (maximum)	Allowable warpage (maximum)
			Teeth per row	Rows per inch		
	A	B				
Inches	*Inches*	*Inch*			*Inch*	*Degrees*
Style A. bastard-cut						
8	13/16	9/32	7	6-2/3	0.020	3
10	1	11/32	7	5-3/4	.020	3
12	1- 3/16	13/32	7	5-1/4	.020	3
14	1- 3/8	15/32	7	4-3/4	.030	3

TABLE XXXIV. *Type XXVII, rasps, wood, class 2, half-round, style A*

Length	Dimensions at largest section		Teeth				Allowable warpage (maximum)
			Back		Flat		
			Teeth per row	Rows per inch	Teeth per row	Rows per inch	
	A	B					
Inches	*Inches*	*Inch*					*Degrees*
Style A. bastard-cut							
6	5/8	15/64	7	9	6	9	4
8	13/16	5/16	8	6-2/8	7	6-2/3	4
10	1	13/32	8	5-3/4	7	5-3/4	4
12	1- 3/16	15/32	8	5-1/4	7	5-1/4	4
14	1- 3/8	17/32	8 to 10	4-3/4	7	4-3/4	4
16	1- 9/16	9/16	8 to 10	4-1/2	7	4-1/2	4

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:39Z
Check the source to verify that this is the current version before use.

3.39 **Workmanship.** Workmanship of the files and rasps shall be of the quality prevailing among manufacturers normally producing files and rasps of the types and classes listed herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Qualification.**[1] Qualification tests shall be conducted at a laboratory satisfactory to the Bureau of Ships.

4.2.1 *Qualification tests.*

4.2.1.1 *Samples for qualification.* The following number of sample files and rasps shall be submitted for qualification tests. These samples will be retained by the laboratory after the completion of the inspection:

1 dozen, 12-inch, type VII, style A, files.
1 dozen, 12-inch, type IX, style A, files.
1 dozen, 12-inch, type XIII, class 2, style A, files.

In addition, the following representative files and rasps shown in table XXXV and which are produced by the manufacturer, are required in connection with the qualification inspection. Should these representative samples satisfactorily pass the qualification inspection, approval will be granted for all files and rasps covered by this specification.

[1] Application for qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3 and 6.3.1).

TABLE XXXV. *Representative files and rasps*

Type	Description	Class	Style	Length	Total number required
				Inches	
I	Aluminum	1, 2	----------	10	2
II	Bandsaw, blunt	1, 2	----------	6	2
IV	Cantsaw	----------	----------	8	1
V	Crosscut	----------	----------	8	1
VI	Double-ender	----------	----------	8	1
VII	Flat	----------	A, B, C, D	10	4
IX	Half-round	----------	A, B, C	10	8
X	Hand	1	----------	12	1
		2	A, B, C	10	8
XI	Knife	----------	A, B, C	6	8
XII	Lead-float	1, 2	----------	10	2
XIII	Mill	1	----------	8	1
		2	A, B, C	8	8
		3	----------	8	1
XIV	Pillar	----------	A, B, C	8	8
XVI	Round	1	A, B, C	8	8
		2 (5/16" dia.)	----------	8	1
XVII	Square	----------	A, B, C	10	8
XVIII	Taper	1, 2, 3, 4	----------	6	4
XIX	Three-square	----------	A, B, C	8	8
XX	Warding	----------	A, B, C	6	8
XXIV	Cabinet	----------	B, C	10	2
XXV	Horse	1, 3, 4	----------	14	8
		2	----------	18	1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:39Z
Check the source to verify that this is the current version before use.

TABLE XXXV. *Representative files and rasps (continued)*

Type	Description	Class	Style	Length	Total number required
				Inches	
XXVI	Shoe, half-round	----------	----------	9	1
XXVII	Wood	1, 2	A, C	12	4

Notes:
1. When a representative length is not available, the next shorter or longer length shall be submitted.
2. When cuts are specified for a particular type and class, the representative length of all cuts shall be submitted.
3. Substitutions of samples other than those shown above shall be agreed to in writing by the Bureau of Ships, prior to authorization of a qualification test.

4.2.1.2 *Qualification tests.* Qualification tests shall consist of the tests as specified in 4.5.2 through 4.5.7.

4.3 Sampling for quality conformance.

4.3.1 *Lot.* All files and rasps of the same type, style, and size presented at one time shall be considered a lot for purposes of quality conformance inspection.

4.3.2 *Sample for quality conformance.* A random sample shall be selected from each lot in conformance with MIL-STD-105 at inspection level S-3.

4.4 Classification of inspection. Inspection of the files and rasps shall be classified as follows:

(a) Qualification inspection.
(b) Quality conformance inspection.

4.4.1 *Quality conformance inspection.* Quality conformance shall be performed on each sample file or rasp selected in accordance with 4.3.2. Quality conformance inspection shall consist of the following.

(a) Examination.
(b) Tests.
(c) Inspection of preservation and packaging.

4.4.1.1 *Examination.* Each file or rasp, selected in accordance with 4.3.2, shall be examined to determine compliance with this specification. Examination shall be conducted in accordance with table XXXVI; acceptable quality level (AQL) to be as indicated. Nonconforming files shall not be offered for delivery, and if the nonconforming files in any one sample exceed the acceptance number for that sample, the lot represented shall be rejected.

TABLE XXXVI. *Classification of defects in accordance with MIL-STD-105*

Categories	Defects
Critical	None Listed.
Major	AQL — 2.5 percent defective.
101	Lot not acceptable, not manufacturer's type number designation as listed on Qualified Products List.
102	Any characteristic not in accordance with specified requirements (type, class, cut number); not as specified.
103	Teeth malformed, missing, broken, or otherwise defective, affecting serviceability.
104	File length (cutting surface) not as specified.
105	Not double-cut where specified.
106	Not single-cut where specified.
107	Not safe where specified.
108	Not parallel as specified.
109	Tang not suitably formed or type specified.
Minor	AQL — 4.0 percent defective.
201	Marking not as specified.

4.4.2 *Tests for quality conformance.* Each sample selected in accordance with 4.3.2, shall be tested for compliance with the requirements of this specification. AQL shall be 2.5 percent defective. Nonconforming files and rasps shall not be offered for delivery, and if the number of defective units exceeds the acceptable number for the samples tested, the lot represented shall be rejected. Quality conformance tests shall

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:39Z
Check the source to verify that this is the current version before use.

consist of the tests specified in 4.5.1 through 4.5.6.

4.5 Test procedures.

4.5.1 *Tang hardness.* Each of the sample files or rasps having tangs shall be tested to determine that tangs are hardened to Rockwell hardness of not more than 30 nor less than 26 Rockwell C.

4.5.2 *Determination of distortion.* Each of the sample files or rasps shall be tested as follows: Distortion for which maximum numerical values are specified shall be measured by means of a dial surface gage and steel blocks supported on a master flat surface (see fig 34), or in some other suitable manner. Flat blocks shall be employed for flat surface files and V-blocks for round and curved surface files. The flat or V-blocks shall be placed under the point and the heel of the file under test and dial gage readings taken at the center on opposite sides by rotating the file through 180°. Minor fluctuations in the measurement, because of file teeth, shall be disregarded. The distortion shall be considered as a deviation from the centerline of the file from a straight line and is one-half of the difference between the maximum and the minimum gage readings expressed in thousandths of an inch. Where the distortion requirement is specified in terms of the projected cross section at the largest dimensions, the file shall be applied to a flat surface and observation shall be made of the tapered section to determine conformity with the applicable requirement of distortion.

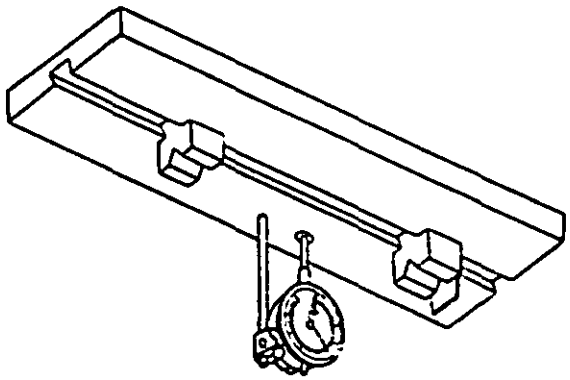


FIGURE 34. Distortion tester.

4.5.3 *Measurement of warpage.* Each of the sample files or rasps shall be tested as follows: Warpage for which maximum numerical values are specified shall be measured by means of an instrument consisting of one horizontal surface, and one pivoted surface, to which an indicating pointer is attached, supported on a master flat surface (see fig. 35), or in some other suitable manner. The horizontal surface shall be adjustable as to position, to accommodate the length of file under test. The point of the file shall be pressed against the pivoted surface, and the heel of the file shall be pressed against the horizontal surface. The warpage shall be indicated by a pointer on a scale graduated in degrees.

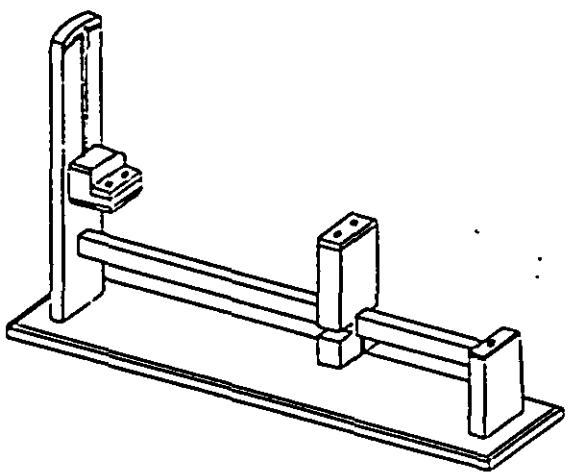


FIGURE 35. Warpage tester.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:39Z
Check the source to verify that this is the current version before use.

4.5.4 *Uniform cutting test.* Each of the sample files or rasps shall be stroked with uniform pressure across the longitudinal surface of a 1/2-inch-nominal-diameter steel rod. At least one stroke shall be made with each cut side and edge of other files and rasps. The teeth along the cutting path shall uniformly cut (bite), and there shall be no indication that tops of the teeth are of unequal height. For this purpose the 1/2-inch rod shall be of steel No. C1020 of QQ-S-634, or its equal approximately, and in an "as-cold-finished condition".

4.5.5 *Test for hardening cracks.* Each of the sample files or rasps shall be tested as follows: The point or tang of the file or rasp shall be held lightly (not gripped), and struck at least one medium light blow with the heel or point of the file or rasp upon an unhardened steel block having a weight of at least 10 times that of the file or rasp. Each sample file or rasp shall pass this test satisfactorily, that is, no fracture shall occur nor shall there be any indication of a crack as evidenced by a deadened tone of the file or rasp when struck upon the steel block.

4.5.6 *Hand prover test.* Each of the sample files or rasps shall be placed upon a wooden-top bench or other suitable location. The hand prover used for this test should be approximately 3/16 inch thick by 1 inch wide and 6 or 8 inches long, and shall be of file steel having a hardness of not less than the minimum allowed in table XXXVII for the applicable file or rasp. One end of the prover shall have a beveled surface with a finish and shape in keeping with the coarseness and shape of the file or rasp to be tested. Care shall be taken to maintain these proper conditions of shape and beveled surface at all times. While the tang of a file or rasp is held down with one hand, the prover shall be held in the other hand with the beveled end of the prover placed near the point of the file or rasp. While holding the prover at an angle so that the beveled end corresponds to the surface of the file or rasp to be tested, it shall be drawn toward the heel of the file or rasp using a steady stroke of uniform pressure not to exceed the maximum allowed in table XXXVIII. Each file or rasp of the sample shall perform satisfactorily in this test, that is, throughout the stroke of the prover, the file or rasp shall act upon the prover similar to the action obtained when using a light pressure in ordinary filing of unhardened steel. Any file or rasp which allows the prover to slide over any portion of its surface shall be considered as unsatisfactory performance. Slide, as used in this instance, means absence of noticeable abrading, cutting, or filing action on the prover.

TABLE XXXVII. *Hardness of hand provers used in test of files and rasps*

Length of files and rasps	Type of files and rasps	Hardness of hand prover (minimum)
		Rockwell C No.
8 inches and under	Bandsaw, double-ender, mill, and taper files.	58
All lengths	Aluminum, flat, half-round, hand, knife, long angle lathe, pillar, round (except chain saw), square, three-square, and warding files.	54
All lengths	Horse rasps.	52
All lengths	Cantsaw, crosscut, and chain saw files.	58
Over 8 inches	Double-ender mill, and taper files.	55
All lengths	Lead-float files; cabinet, shoe, and wood rasps; and milled, curved teeth files.	47

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:39Z
Check the source to verify that this is the current version before use.

TABLE XXXVIII. *Pressure exerted on hand provers in test of files and rasps*

Length of files and rasps	Shapes and sections of files and rasps	Pressure on prover (maximum)
		Pounds
Ten inches and longer	Flat sides	10
Under 10 inches	Flat sides	8
Ten inches and longer	Round	7
All lengths	Half-round	7
Under 10 inches	Round	6
All lengths	Knife file edges and saw file edges	4

4.5.7 *Machine-cutting tests.* Upon satisfactory completion of the tests for distortion and warpage (see 4.5.2 and 4.5.3), machine-cutting tests of the 12-inch flat bastard, half-round bastard, and mill regular bastard files shall be conducted, using hacksaw machines adapted for file cutting tests. The files shall be clamped in the frame of the machine and operated for 20,000 strokes on each side of 3 mill and 3 flat bastard files and the convex side only of 3 half-round bastard files. The effective length of each stroke made by the hacksaw machine shall be 6 inches. The files shall cut 1-inch square, class 3 carbon tool steel (carbon 0.81 — 0.90 percent; manganese 0.15 — 0.35 percent). The steel shall be annealed, hardened, and drawn to a Rockwell hardness number of between C-24 and C-30. The files shall be operated at 60 strokes per minute under a pressure at midstroke of 30 pounds for the flat and half-round, and 25 pounds for the mill files (see table XXXIX).

TABLE XXXIX. *Machine cutting test requirements*

Type of file	Cut	Load at midstroke	Length of bar removed in 20,000 strokes by one side of file	
			Minimum average[1]	Minimum individual
		Pounds	*Inches*	*Inches*
VII—Flat	Bastard	30	1.50	1.10
IX—Half-round	Bastard	30	1.00	0.75
XIII—Mill, class 2 regular	Bastard	25	0.50	.35

[1] However, the sum of the average amounts of bar removed by types VII, IX, and XIII files tested shall be not less than 8.50 inches.

5. PREPARATION FOR DELIVERY

5.1 **Preservation, packaging, packing, and marking.** Preservation, packaging, and packing shall be in accordance with MIL-H-15424. Levels of preservation and packaging shall be as specified (see 6.2).

5.2 Marking.

5.2.1 *Civil agencies.* In addition to marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military activities.* In addition to marking required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:39Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use.

6.1.1 Type I, aluminum files are primarily for filing aluminum, aluminum alloy, and may be used on other nonferrous metals. Teeth are designed for fast cutting and to minimize clogging.

6.1.2 Type II, bandsaw blunt files are for sharpening bandsaw blades having well rounded gullets.

6.1.3 Type IV, cantsaw files are used for sharpening crosscut saws with "M" teeth and circular saws and may be used for sharpening saw teeth with less than 60° angle.

6.1.4 Type V, crosscut saw files are for sharpening crosscut saws of the "Great American" style; the rounded back is used for deepening rounded gullets of this style saw.

6.1.5 Type VI, double-ender and type XVIII, taper files are for sharpening standard hand saws and may be used for sharpening other saw teeth having a 60° angle.

6.1.6 Type VII, flat files are used by machinists, millwrights, and other craftsmen for general purpose filing.

6.1.7 Type IX, half-round files are for general purpose filing and as they will file concave as well as flat surfaces, are suitable for carrying by maintenance men.

6.1.8 Type X, hand files are used for flat surfaces and having one safe edge, are ideal for filing close to shoulders.

6.1.9 Type XI, knife files are used primarily by tool and diemakers on work having acute angles and are suitable for keyways or slots. The back is safe (uncut).

6.1.10 Type XII, lead-float files are used for filing lead, babbitt, and other soft metals.

6.1.11 Type XIII, class 1, mill files are used for general purpose filing by craftsmen preferring parallel sides on the files.

6.1.11.1 Class 2, mill files are used for sharpening mill saws and various cutting tools and as a general purpose shop file.

6.1.11.2 Class 8, mill files are used primarily for sharpening mill saws as they have two rounded edges for filing the gullets and also may be used for sharpening other tools.

6.1.12 Type XIV, pillar files are used by millwrights and mechanics for enlarging keyways, slots, etc., and have one safe edge.

6.1.13 Type XVI, round, class 1, regular files are used to enlarge circular openings, elongate holes, and similar purposes. Class 2, chain saw files are for sharpening chain saws.

6.1.14 Type XVII, square files are used to enlarge square holes, filing slots, keyways, and similar purposes.

6.1.15 Type XVIII, taper files are for sharpening standard hand saws or other saws having 60° angle teeth. In general, the extra-slim and double-extra-slim in the shorter lengths are for fine teeth saws and the longer lengths of the regular or slim are for coarse teeth saws.

6.1.16 Type XIX, three-square files are for filing internal angles, clearing out corners, etc., and may be used for filing taps and cutters in the annealed state.

6.1.17 Type XX, warding files are used by locksmiths for filing ward notches in keys and also by other mechanics for filing narrow slots and in close spaces.

6.1.18 Type XXII, long angle lathe files are used for obtaining smoother finishes on lathe work and uncut edges permit working close to shoulder without damage to it.

6.1.19 Type XXIII, milled curved teeth files are used primarily for finishing large flat, or slightly curved surfaces of materials in the soft categories but may be used on mild steel.

6.1.20 Type XXIV, cabinet rasps are used

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:39Z
Check the source to verify that this is the current version before use.

by cabinetmakers and woodworkers, and as the teeth are finer than those on half-round wood rasps, they produce a better finish.

6.1.21 Type XXV, horse rasps are used in connection with shoeing horses. The rasp teeth on one side are for fast cutting in shaping the hoof; the file teeth on the opposite side are for filing the nails, shoes, and general finishing.

6.1.22 Type XXVI, shoe rasps were originally used by shoemakers and leather craftsmen but have been adopted by cabinet, pattern, and modelmakers for general use on soft materials. Each side has half its length cut with rasp teeth and the other half file teeth.

6.1.23 Type XXVII, wood rasps are generally used by wood craftsmen, wheelwrights and plumbers, for shaping wood items, enlarging openings.

6.2 Purchasers should select the preferred options permitted herein and include the following data in procurement documents:

(*a*) Title number, and date of this specification.
(*b*) Type, class, and style required (see 1.2).
(*c*) Length of file or rasp required (see 3.9 and applicable table).
(*d*) If type XIII files are to be other than single-cut (see 3.26).
(*e*) Selection of applicable levels of preservation, packaging, packing, and marking required (see 5.1).

6.3 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion on Qualified Products List QPL GGG-F-325, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Goverment tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Bureau of Ships Department of the Navy, Washington D.C., 20360, and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Ave., Philadelphia, Penn. 19120.

6.4 This specification conforms with Simplified Practice Recommendation R6-63 insofar as is practicable.

6.5 The following types, classes, and styles are obsolete and are no longer listed in R6-63:

Type III—Brass, half-round.
Type VIII—Foundry, classes 1 and 2.
Type X—Hand, class 1, finish.
Type XV—Pitsaw.
Type XXI—Auger-bit.
Type XXIV—Rasp, cabinet, style C, smooth-cut.
Type XXVII—Rasp, wood, class 1, flat, style C, smooth-cut.
Type XXVII—Rasp, wood, class 2, half-round Style C, smooth-cut.

MILIARY CUSTODIANS:
Army—GL
Navy—SH
Air Force—69

Review Activities:
Navy—SH

Preparing Activity:
Navy—SH

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 20 cents.

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:39Z
Check the source to verify that this is the current version before use.